SVG-DOM-GRAPH / DESIGN SPECIFICATION

Discrete-event service simulator Gallery demo panel

Revision 4 | 10 November 2026 | Pre-implementation specification

Disposition. Revision 4 incorporates the three final clarifications from the revision 3 review, while retaining the accepted S0-S5 baseline and the D1-D5 decisions. It fixes the known-end Play dispatch, isolates the shared JavaScript detach injector and defines its self-test. P0-P4 and their exit gates are unchanged. No new gallery implementation or pass is claimed.

Executive decision

Place the simulator immediately after Petri Net in the Concurrency group. Keep each compatible SVG scene long-lived, update the live eight-cell data node on every committed slice, and use the existing controller for playback, seeking and repair. The new user path must be the real gallery route, not the S5 test fixture.

Decisions settled for revision 4

Decision	Accepted rule
D1 - Catalogue	E0, E3-NOOP, scaled mixed_outage/7 (360 arrivals), and advanced W6 (4,000 arrivals). W6 is **not** a benchmark path.
D2 - Seek maximum	One preflight replay with deterministic per-catalogue slice/pop ceilings and exact expected totals; elapsed time is diagnostic only.
D3 - Staging	P0 first tests a connected **opacity:0 + inert** candidate; a measurable off-screen inert host is the fallback. Presence, focus and AX exclusion are gates.
D4 - Keyboard	Native HTML controls only; no wrapper-level or document-wide shortcuts in this release.
D5 - Browser scope	Chrome automated support; log Firefox and Safari observations separately, without claiming parity.

Release constraint: The cross-size transaction and observer indirection belong in **service-sim**. Real-gallery failures use a page-level JavaScript injector, never a second wasm build with `fault-hook`.

1. Baseline and scope

The governing engine specification remains **design-specs/discrete_event_service_simulator.pdf**, revision 6 (especially section 9). Stages S0-S5 have implementation and test evidence; this PDF is an integration contract, not a re-approval of their algorithms.

Existing component	Inherited contract	New gallery obligation
S1: service-sim-engine	Integer-tick deterministic engine; independent JS oracle; known-answer and golden fixtures.	Use existing factories, fixed revision/seed; no new engine in the page.
S2: service-sim Surface	Fixed scene, 1-4 servers, 12 queue windows, ten overlay aggregates, eight u64 data cells.	Same-size navigation preserves element identity; one setter call per committed draw.
S3: Controller and browser host	Committed cursor; async seek; epochs/tokens; hidden/detach guards.	Bind native controls; one stable lifecycle owner, with an indirect current-controller target.
S4: recovery	Dirty redraw, torn setter recovery, W5 ten-point fault executor.	Expose Repair display; cross-size rollback must preserve the old picture.
S5: W6	Actual product controller behind test-only controls; Layer A/B/C evidence.	A real gallery route; do not reclassify test fixture timings as gallery timings.

What is deliberately out of scope

- No new event ordering, PRNG, hashes, checkpoints, accelerated playback scheduler, 100k-pop browser UI or new data-node type.
- No alternate gallery-owned simulation state machine; the adapter cannot shadow a committed cursor, dirty state or pending seek.
- No `prepare_ahead()` in ordinary gallery stepping, seeks or playback: that API is reserved for W6 Layer B precomputation; no production `fault-hook`.

Concrete integration files

`demo/panels/16-concurrency/02-service-simulator.html` and a new `demo-app/src/service_sim/` module; register `panel-service-simulator` through `demo_gallery!` so the generated menu, lazy init and source frame stay aligned. The new reusable coordinator lives in `service-sim/src/`.

2. Scenario catalogue and bounded seek extent

User-facing label	Deterministic source	Known extent
Walkthrough - default	E0; 2 servers, queue capacity 3.	12 slices; 13 snapshots (0..12).
No-op and stale events	E3-NOOP; 1 server, capacity 2.	9 slices; 10 snapshots (0..9).
Busy service / outages	**Scaled run** of frozen mixed_outage/7; seed 20261106; 360 arrivals, 3 servers.	1,009 pops; 487 slices (0..487).
W6 reference trace - advanced	Same frozen mixed_outage/7 revision and seed; 4,000 arrivals, 3 servers.	11,214 pops; 5,464 slices (0..5464). Not a performance benchmark.

Terminology: The 360-arrival run is scaled from the existing frozen generator revision; it is **not** a new frozen revision. A snapshot index includes the initial state zero, while a slice is a successful simulation tick advance.

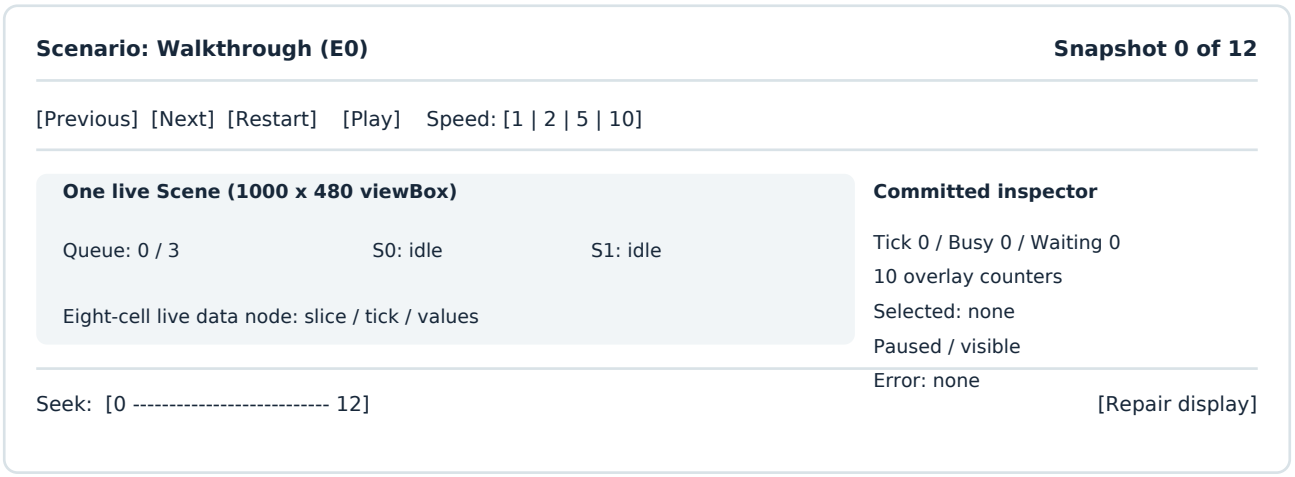
D2: exact maximum through bounded preflight

- On catalogue selection, replay a separate candidate engine to natural completion without writing a Surface, checking the deterministic slice/pop bounds throughout. Do not instantiate a live controller until successful preflight.
- Compare preflight count and pop total with the corresponding fixture or manifest record. Disagreement or limit/overflow is a failed preflight: retain the active choice and explain the mismatch.
- Bound work **deterministically** by per-catalogue maximum slices and event pops: E0 (12 / 19), E3-NOOP (9 / 9), 360-arrival scale (487 / 1,009) and W6 (5,464 / 11,214). Refuse overflow, exceeding either ceiling, or final count mismatch. Time taken is logged only; **never** a reason to refuse a valid run. Cache by scenario id, revision, seed, arrivals and all parameters.
- The slider maximum is the preflight end even when `Controller::last_slice()` is None. A seek clamp or failure resynchronises the control to the committed index. The preflight never invokes `prepare_ahead()` on the live controller.

The preflight is outside timed drawing. For W6, the complete range 0..5464 is intentional. The advanced path has no automatic 5,464-step playback and no W6 metric reporting.

3. Panel composition

Use native HTML transport controls outside the zoomable SVG; the internal scene has the existing fixed **1000 x 480** viewBox. The drawing below is structural and not a pixel layout.



One toggle, not two buttons: PageHost already synchronises a single Play/Pause control, setting its text and `aria-pressed` from actual controller playback state. The panel therefore supplies exactly one toggle button and one listener.

Region	DOM and visual contract
Header and instructions	What arrivals, priority, server outage, timeouts, retries and cancellations mean; readable legend with non-colour state words.
Transport	Native Previous, Next, Restart, one Play/Pause toggle, a labelled speed select (1, 2, 5, 10 slices/s) and Repair display.
Seek	Labelled range with min=0, preflight maximum and separate requested/committed text. No speculative displayed commit.
Scene	One SVG per compatible server count, fixed connectors and queue slots, one overlay and one eight-cell live data node.
Inspector and voice	HTML inspector reads only committed_model(); visible status, one polite live region and actionable error area. No second simulation.

Scene persistence contract

Next, Previous, Restart, Seek, selection and same-size scenario changes must not replace the SVG, `svg-dom-graph-content` group, nodes, overlay slots or data cells. A cross-size switch is the sole allowed scene replacement, and only after a successful staged publication.

4. Controls, boundaries and presentation

Action	Dispatch and result
Next / Previous	`next()` / `previous()`; one slice, not a pop. Previous may be Pending until its frame commits.
Restart	`restart()`; at index 0 still reaches the controller, invalidates previous navigation and repairs a dirty page.
Seek	Each distinct range `input` value calls `seek(index)`, including one replay per arrow-key press; a terminal change must not duplicate the same request. Last-wins is deliberate.
Single Play/Pause toggle	If playing, `pause()`. Otherwise, at the preflight-known end call `next()` for a synchronous boundary/repair with **no timer**; at earlier indices call `set_speed(ms)` then `play()`. Always show Play / aria-pressed=false at the end.
Rate	1/2/5/10 slices/s -> 1000/500/200/100 ms, via `set_speed()`. In-flight timer replacement is already a controller concern.
Scenario	Preflight; then in-place switch or coordinated replacement. Only successful commit updates chosen option, current index, maximum and inspector.
Repair display	`repair()`. Restore the committed picture and control ARIA; do not change cursor, selection, view, scenario or resume playback.
Selection	`select_job(Some(id))` / `select_job(None)`; publish inspector only after the selection draw commits.

Boundary controls are still real requests

> **Mandatory:** When a direction has nowhere to go, render the button with `aria-disabled="true"` and a visible/accessible reason such as “Already at first snapshot”. **Never use the HTML** `disabled` **attribute.** Its click handler must still dispatch to the controller. An `AtBoundary` outcome may repair a dirty page and must not be swallowed.

Only committed state updates the inspector and data display. `Outcome::{Done, AtBoundary, Pending, Unavailable, Failed}` stay separate; an unavailable host is not a failed setter. A pending seek presents both requested and committed indices without inventing a commit.

Seek accessibility and Play at natural end

Give the range `aria-valuetext="Snapshot 8 of 12"` (using requested position while seeking) and a description naming the still-committed snapshot. Native range arrows each dispatch a new seek; do not debounce or override the controller last-wins token. A test must exercise multiple rapid arrows on W6.

At the **preflight-known final snapshot**, the Play-toggle handler calls `controller.next()` **instead of** `controller.play()`. A naturally completed engine returns `AtBoundary` synchronously: `step_next()` records `last_slice` even when it was previously None, and the boundary path repairs a dirty display before returning. **No timer is scheduled and the toggle never transiently reads Pause.** Show **End of scenario**; keep Play / `aria-pressed="false"`, cursor and data node unchanged. A failed boundary repair reports the failure rather than declaring success. At any earlier committed index, Play still calls `set_speed()` then `play()`.

Keyboard scope (D4)

Native buttons, select and range use their built-in Tab, Space, Enter and arrow-key behaviours. No global keyboard map, and no diagram-wrapper shortcuts in this release. Scene job selection retains the existing SVG focus/navigation behaviour only where already provided.

5. Data vocabulary and inspector contract

The engine stores **12 canonical counters**. The overlay intentionally projects **ten displayed aggregates**. The live SVG data node has **eight u64 cells**, of which cell 0 is the slice ordinal. These are three distinct structures; tests G04 and G17 must not compare them as though their lengths or meanings coincide.

Ten overlay counters	Committed source value
Arrived	`c.arrivals`
Admitted	`c.admitted`
Rejected	`c.rejected`
Completed	`c.completed`
Timed out	`c.timed_out`
Cancelled	`c.cancelled`
Failed	`c.failed_attempts + c.failed_deferrals`
Waiting to retry	`model.retry_wait`
Stale pops	`c.stale`
No-op pops	`c.noops`

Canonical engine counters - not a second HTML dashboard

`pops, stale, noops, slices, arrivals, admitted, rejected, completed, timedOut, cancelled, failedAttempts, failedDeferrals`. These 12 remain available to the independent state oracle and checkpoint hash. The gallery inspector must not invent a different “12-counter” interpretation.

Inspector and eight data cells

HTML inspector: committed snapshot index, simulated tick, waiting/queue capacity, busy servers, selected job details (or none), and the ten named overlay counters above. Source: `Controller::committed_model()` plus committed selection. The single SVG data node is written by `Surface` as **[snapshot, tick, waiting, busy, completed, timed-out, cancelled, rejected]**, with u64 decimal readback. It is not the ten-counter overlay.

S2 requires one data setter on every successful visible draw, including a slice made only of stale pops. A failed draw is never counted as a committed slice, and a full repair may legitimately call the setter again without advancing the ordinal.

6. Ownership: one stable panel target

A gallery panel survives route changes. A cross-size switch replaces its controller and SVG, while `GalleryHint::install` and `PageLifecycle::install` capture the callable target when installed. Capturing the original controller directly would leave both hooks calling a disposed one after the replacement. Revision 2 prohibits that.

Stable lifetime owner (new integration contract)

Object	Owner / lifetime / responsibility
`PanelOwner`	One retained per initialized gallery panel. Holds `active: Option<Controller<PageHost, BrowserScheduler>>`, scenario metadata, controls, one GalleryHint, one PageLifecycle and listener handles.
Observer and lifecycle closures	Installed exactly once and capture a **Weak** reference to PanelOwner. On notification, obtain the current controller from the owner and call `presence_changed()` or `suspend()`.
Controller target swap	The coordinator publishes the new controller to the stable owner only when a staged cross-size transaction commits. On rollback the owner still points to the old controller.
Scheduled callbacks	Controller keeps its existing weak target and generation/epoch/token checks. They cannot publish a cancelled or disposed commit.
Disposal	Unregister host listener handles, drop GalleryHint and PageLifecycle, dispose active controller; drop owner. No strong callback -> owner reference cycle.

The adapter must clone a controller handle from the owner under a short borrow and **release the owner borrow before invoking it**, preventing synchronous callback reentrancy through `RefCell`. A disappearing owner is a no-op, not a fresh initialization.

Notification ordering

- Before a swap, notifications target the old active session; a staged session is never observed as committed.
- At publication, update the active handle, current host and cached activation as one coordinator-owned commit boundary; then synchronise visible controls and inspector.
- The gallery router remains the existing `#stage` / `.section.active` path. Do not register observers on each scenario or route revisit.

7. New cross-size transaction boundary

The cross-size rebuild is simulator session management, not a gallery-specific algorithm. Implement it in **service-sim** behind an explicit **surface/session factory** and a publication interface that can be faked natively. The gallery supplies DOM host creation and removal, focus targets, catalogue metadata and the stable owner; it does **not** decide commit/rollback semantics.

Suggested responsibility split (names illustrative)

service-sim coordinator	Gallery implementation
Own the state machine: Idle -> Staging -> Publishing -> Committed or RolledBack. Stage a candidate session; check `Outcome`, `is_dirty()`, host presence and shape; decide rollback.	Create an SVG with unique id and confirmed sizing; provide callbacks to hide/show/restore hosts and to preserve focus. No engine manipulation.
Surface factory / publication adapter traits are abstract over candidate creation and commit/undo; fake implementations enable fast native tests with held failures.	Bind production factory to `Surface::build`, PageHost and BrowserScheduler. Update selector only on coordinator success.
Retain the old live session until publication fully succeeds; cancel/invalidate its pending work at request ownership boundary.	Keep one observer/lifecycle owner through both staging and committed state; no duplicate listeners.

Ordered protocol

- **Preflight:** check parameters, exact seek extent, resource budgets, host live status; record old cursor, scenario, speed, view and focus. Reject without allocating when invalid.
- **Stage:** build a uniquely named, connected, measurable and accessibility-excluded host within the active panel (only after P0 spike passes). Construct new Surface/PageHost/Controller at snapshot 0; apply saved speed while paused.
- **Validate:** `Controller::new` returning 0k is insufficient: the initial draw may leave `is_dirty()` true. Confirm clean projection, actual data-node readback, presence and no leaked/duplicate IDs.
- **Publish:** stop and invalidate the old playback/seek; perform host visibility and pointer/focus handoff with compensating undo; update stable owner pointer and metadata only after the new scene is fully live and verified.
- **Rollback:** if any stage, draw, focus or DOM publication operation fails, restore old host and active pointer, dispose/remove the candidate and repair the prior committed picture if needed. The old simulation cursor, selection and scenario remain unchanged.
- **Finalise:** only after publication succeeds, dispose the old controller and remove its SVG. Preserve user speed preference; do not automatically resume Play.

> **Important distinction:** A scenario-switch request is allowed to stop the old playback, even if publication later fails, because request ownership invalidates old timers. The rollback guarantee preserves **committed simulation data, view, selection and scenario**, not the formerly running timer. Do not promise restoration of a cancelled timer.

8. P0 gate: prove staging before building the panel

D3 is no longer a late design note. The first implementation work is a dedicated browser spike that either proves a safe connected staging host or settles the off-screen fallback. No gallery controls or cross-size replacement are built until this gate is recorded.

Probe	Pass condition / observation
P0-S1: host presence	First candidate: a connected, sized SVG or staging container in the **active** section with `opacity:0` and the native `inert` property (pointer-events:none may supplement it). Check `isConnected`, default `checkVisibility()` and `PageHost::presence_for_test()`.
P0-S2: geometry	Construct actual `Surface::build` into it. Check viewBox and measurable scene/labels, finite geometry and no overlap/overflow introduced by staging. Exercise initial full draw and data setter.
P0-S3: accessibility	With `inert`, test Chrome AX-tree exclusion, sequential Tab, scripted focus, and pointer interaction; none may target the candidate. `aria-hidden` alone is explicitly insufficient for focus isolation.
P0-S4: publish / rollback	Switch CSS/pointer/accessibility state to active and back after injected failures. At every checkpoint there is exactly one exposed scene and a valid active target.
P0-S5: fallback	If opacity + inert fails, repeat with a connected **off-screen, inert** SVG that retains non-zero layout geometry. Check clipping, finite measurements, focus/AX exclusion and visibility predicate; never construct under display:none or visibility:hidden.

Chrome is the release gate, with Firefox and Safari observations logged separately. Neither opacity nor `aria-hidden` guarantees inert focus behaviour: the spike must confirm that `inert` preserves measurable construction and an independently verified live presence predicate.

P0-F: exercise gallery faults without a second wasm build

Before P1, load `perf/page/fault-injector.js` through CDP before the real gallery scripts. Reuse its unchanged `__faults`, `__timers` and `__frames` APIs. Add a separately armed **test-page-only detach-on-write mode** (proposed interface: `__faults.armDetach({svgId, method, nth})` and `disarmDetach()`). It starts **off**, is independent of exception arming, and must not change which calls, or how many calls, the existing recorder and exception injector count. Do not arm detach and exception failure for the same operation.

When armed, first count only the requested eligible mutations, preserve the ordinary wrapper call-count path, perform the native write, then synchronously detach the selected SVG before returning to wasm. The report must name the wrapped method, ordinal and exact firing site; missing/duplicate firing fails G14. Test against a live `Surface` draw (overlay and data-node attributes), reattach for repair, and ensure the untouched normal production build never installs or enables a Rust `fault-hook`.

P0-F self-test - compatibility of the shared injector

- With the detach mode unarmed, replay `selftest-recovery.mjs` and the existing recorder, DOMException, held-timer and held-frame assertions unchanged; recording/counting must match its prior vectors.
- Arm at a known wrapped write and assert exactly one detach after that native operation; verify a too-large ordinal reports **not fired** and fails the test rather than passing vacuously.
- Disarm and repeat the original call trace; method/ordinal counts are identical to the no-detach trace. The new mode leaks no arm state, wrapper, timer, frame or listener after cleanup.

P0 deliverable

Check in the staging and injector spikes, browser version, CSS, measured geometry/presence/AX/focus results, armed/fired mutation record and explicit pass/fail. If neither staging candidate passes, block cross-size switching. A P0 failure to detach during a real draw is a failed gate, not evidence of recovery.

9. Visibility, lifecycle and repair

The gallery keeps hidden sections connected, toggling `.section.active`. An `isConnected` check alone cannot permit a draw. S0 and S3 already established the real gallery ordering; the panel must exercise it through the stable owner of section 6.

User or host transition	Required outcome
Route away (observer delivers while hidden)	A MutationObserver callback sees the panel hidden, targets the current controller, invalidates playback/seek and preserves the committed picture.
Route back	Call presence_changed on the current controller. Repair if dirty. Remain paused and show “Paused while hidden - press Play to resume” if interrupted.
Document visibilitychange events	An actual hidden transition dispatches `visibilitychange`, which stops Play; a later visible event must not resume it, even within a fast sequence.
pagehide / pageshow	pagehide calls suspend unconditionally; pageshow checks/repairs but never resumes.
Same-task gallery class hide then show	One MutationObserver batch can observe only the final **live** state, so cancellation is **not** guaranteed. Continuation is permitted; assert no double tick, no duplicate callback/observer and correct active owner.
Detached before or during draw	No success commit; cease further writes to detached root; later return requires full repair before incremental updates.
Dispose panel owner	Remove observers, lifecycle handlers and DOM listeners; dispose controller; kept callbacks are inert.

Failure presentation

- Use `Outcome::Unavailable` for hidden/detached/disposed, and `Outcome::Failed` for write/scheduler/engine errors; do not conflate them.
- Report a recoverable write problem once per failure episode in a visible error region. Repair display must restore committed cells, overlay and Play ARIA, without committing a new slice.
- A boundary click can be a repair request. A failed repair remains a failure even at index zero. Stale callbacks after a hide may execute but must be inert.
- The displayed paused-while-hidden reason is a presentation of a real interruption, not a new controller state; it clears when a user deliberately starts Play or another relevant action.

10. Accessibility and responsive presentation

- Controls are ordinary labelled HTML: one Play/Pause button driven by PageHost, four rate options, scenario select, range, Previous/Next/Restart and Repair display. Focus stays on the control that was activated.
- A boundary button uses `aria-disabled="true"`, remains clickable and keyboard activatable, and includes an accessible reason. It must not use `disabled`.
- Provide one `role="status" aria-live="polite"` region for deliberate changes; do not announce every 10 Hz slice. Use a distinct actionable error region and associate it with Repair display.
- Queue and server states have both visible text and accessible names. Priority, outages, busy, idle, selection and stale/no-op counts are not conveyed by colour alone.
- During staging and after a successful cross-size switch, no two scenes are simultaneously exposed in the accessibility tree. A candidate cannot receive focus; after commit restore a valid focus target.
- At narrow viewports controls reflow outside the SVG. SVG zoom/pan must leave HTML text and controls at usable size; scene view and control focus persist through ordinary navigation.

State messaging examples

State	Visible / accessibility rule
Ready	“Snapshot 0 of 12; simulated tick 0; paused.”
Seeking	“Seeking snapshot 8…” while pending. Inspector still reports last committed index.
Playing	“Playing at 5 slices/s”; toggle text “Pause” and `aria-pressed="true"`.
Boundary	“Already at first snapshot” / “End of scenario”. No alert.
Interrupted by hiding	“Paused while hidden. Press Play to resume.” Announce once per episode.
Failed / unavailable	“Display update failed; snapshot N unchanged. Use Repair display.” or “Simulator is hidden or unavailable.”

11. Acceptance tests: gallery controls (G01-G10)

All G-tests use the real gallery route and controls. The oracle is the existing JS engine or S1 known-answer ledger, not a hand-written state machine in the panel. “CDP” below means one Node script with one Chrome session; deterministic controller checks also run natively where specified.

ID / lane	Pass condition
G01 / CDP	Generated menu places panel immediately after Petri Net; deep-link and source frame agree with the section id and `demo_gallery!` registry.
G02 / CDP	First route builds once; revisits create no second owner, observers, SVG, listener set or duplicate id.
G03 / CDP	E0 initial full draw: actual SVG, queue/servers, one overlay and eight data cells match initial fixture.
G04 / CDP	12 real Next clicks: committed model, ten aggregate labels, eight data cells and projection match E0 known answers. Each successful commit advances data ordinal.
G05 / CDP	Previous, Restart and nonsequential Seek equal full redraw; rapid keyboard arrow inputs on W6 each request a replay, last-wins; AX range announces Snapshot N of M.
G06 / native + CDP held	1/2/5/10 Hz with held timers: one slice per tick and rate replacement. At a preflight-known end, clicking Play calls next(), schedules zero timers and keeps the toggle unpressed, for clean and dirty boundary/repair. One real 1 Hz Play interval only.
G07 / native + CDP held	Held callbacks from Pause, Next and Seek are inert; boundary replacement remains last-wins. Pause does not cancel a pending seek.
G08 / CDP	Repair on clean and dirty pages commits no slice and restores data cells, overlay and Play ARIA.
G09 / CDP	Select and clear a job: committed selection, outline, inspector and accessible label agree; cursor and view unchanged.
G10 / CDP	Native keyboard controls, names, focus retention and boundary `aria-disabled` rule pass Chrome AX inspection.

12. Acceptance tests: lifecycle, scenarios and stress (G11-G20)

ID / lane	Pass condition
G11 / CDP held	Route away with held 1 Hz Play timer. One real gallery MutationObserver signals the current controller; late callback inert; route back paused.
G12 / CDP synthetic	Separate: document visibilitychange hide/show must cancel; gallery class hide+show in one observer batch may continue. In either case no extra/double tick, stale target or observer leak.
G13 / CDP held	Hide during pending Seek with held frame. Replaced frame inert; a later user Seek succeeds.
G14 / native + CDP injected	Single gallery wasm, page-level injector: detach before draw or synchronously after Nth real DOM write mid-draw; prove injection fired, no commit, then full repair. Native fake host covers coordinator boundaries.
G15 / CDP injected	Same-size switch publishes slice 0 atomically; page-injected DOMException retains prior scenario, selection, generation and projection.
G16 / native + CDP injected	2 -> 1 -> 3 -> 2: connected inert candidate, exactly one AX scene; page-injected construction/draw/publication failures and native fake focus failures roll back to old target.
G17 / CDP	E3-NOOP nine slices: exact model/overlay/data readback including stale/no-op figures; no false commit.
G18 / nightly CDP	Scaled mixed_outage/7 487-slice path, spot-check independent state and DOM hashes, random seeks, rate changes and live data node.
G19 / nightly CDP	Advanced 4,000-arrival trace: 1,000 successful gallery commits, sampled seeks and no benchmark classification.
G20 / CDP + native / nightly soak	Repeated cross-size switches, hides/revisits, disposal and retained callbacks return DOM/listener/closure/externref counts to baseline. Short gate per commit, full endurance nightly.

Native coordinator test additions

Add focused fake-host tests for initial dirty candidate, failed factory, same-size path, failed staging, publication failure at each DOM step, failure restoring focus, stable callback retarget after commit, rollback retaining the old target, exactly-once old disposal, no strong ownership cycle, and stale timer/frame after replacement. No browser is needed for these contract-level tests.

13. Deterministic timing and CI placement

Do not enlarge `perf-smoke` or any CDP shard. One Node entrypoint, `dess/gallery/test-gallery.mjs`, serves the **real, prebuilt gallery** in one Chrome session. Run G01-G17 and short G20 in a new `dess-gallery` CI job, **parallel** to perf-smoke and CDP shards, with `needs: cdp-build` to reuse its `cdp-builds` artifact; do not rebuild the demo. Nightly G18/G19/extended G20 run with an explicit nightly flag, separate from W6 Layer C timing.

Lane	Where and when	Timing mechanism
Native / per commit	`cargo test -p service-sim` and the new coordinator fake-host tests.	`ManualScheduler` with kept/cancelled callbacks; zero wall-clock waiting.
CDP gallery / per commit	`dess-gallery`, after cdp-build artifact; G01-G17 and short G20.	Inject the shared page-level fault script before gallery JS. G06/G07/G11/G13 use its held timers/frames on the same production wasm; G14-G16 use real DOM injection.
CDP real-timer smoke / per commit	Inside `dess-gallery`, one 1 Hz Play/Pause and one frame readback.	Real `BrowserScheduler`, no repetition over four speeds; elapsed time is diagnostic only.
Nightly CDP / opt-in	`dess-gallery-nightly` in perf-nightly workflow: G18/G19/expanded G20.	Same real gallery, independent correctness probes; separate from `w6-play` and the per-commit lane.
Other browsers / observations	Manual Firefox and Safari matrix, versions recorded.	No pass claim or CI parity requirement until tested and supported.

Test instrumentation boundary

Instrument **the real production gallery wasm** at page level: load `perf/page/fault-injector.js` before scripts; use `__timers`/`__frames` for held callbacks, `__faults.arm()` for DOM exceptions, and its narrowly added synchronous detach-on-write mode for G14. Never add Rust test hooks or a second gallery build. For each fault, assert the instrumented method and index actually fired; absence is a test failure. Preserve a separate uninstrumented real-timer smoke.

Evidence record: for each G-test capture route, scenario/seed, requested vs committed index, engine hash, DOM projection hash, data ordinal, active generation/epoch/token, host presence, AX-scene count, listener/closure deltas, real vs controlled timer label, browser version and failure injection (if any).

14. Build integration, size and risk

The demo crate currently depends on `svg-dom-graph`, demo-common, svg-dom, spmv components and web bindings. The new panel requires adding the unpublished `service-sim` dependency (which itself brings `service-sim-engine`, including sha2). Treat the compiled wasm size as an explicit product metric, not an unmentioned side effect.

Required check	Evidence / gate
Release feature isolation	CI must run `cargo tree -p svg-dom-graph-demo --target wasm32-unknown-unknown -e features` against the demo build graph and **fail** if `service-sim feature "fault-hook"` appears. Preserve the captured tree; Cargo features can unify across workspace members. Gallery faults come from JS only.
Build impact	Measure the exact before/after demo wasm byte count and gzip size using identical toolchain, `wasm-pack` flags and wasm-opt setting. Record delta in bytes and percentage; do not impose an invented pass threshold.
Dependency audit	Record the pre/post `cargo tree` for the wasm target, including `sha2`; deduplication is a measured fact, not an assumption.
Start-up / lifetime	One lazy panel init; show when selected; revisit without reconstruction. Initial build failures use gallery's existing Result/error path, never a panic.
Scene / memory	No element count growth during same-size draws; no observer/listener/externref residual across true disposal or repeated coordinator replacement.

Key risks and mitigations

- **Staging excluded from AX, pointer and focus but still drawable:** inert + opacity P0 spike with off-screen inert fallback; otherwise block cross-size changes.
- **Hook points at disposed controller:** stable weak PanelOwner indirection with native retarget and rollback tests.
- **Half-published two-host state:** service-sim coordinator with compensated host visibility/focus rollback, plus Chrome AX inspection.
- **Long running CI:** standalone parallel dess-gallery job reusing cdp-build artifact, with held time and separate nightly stress.
- **UI double truth:** commit data and metadata together; selector cannot lead an uncommitted model.

15. Implementation phases and exit gates

Phase	Work and required exit
P0 - staging and injector proof **first**	Prove opacity:0 + inert presence, geometry, Tab/script focus, pointer and AX; off-screen inert fallback if required. Exercise default-off, count-preserving page-level detach; run its firing/missing-firing self-test and existing recovery self-tests without enabling fault-hook.
P1 - reusable session coordinator	Implement surface/session factory and transactional coordinator in service-sim, native fake-host rollback, dirty-candidate and stable-target tests; no simulator semantics in demo-app.
P2 - gallery scaffold and controls	Add fragment/registry/source frame, exact counters, deterministic preflight, seek aria-valuetext and one Play/Pause toggle. G01-G10 pass in dess-gallery.
P3 - lifecycle and switching	Install stable PanelOwner, distinguish event hide from one-batch route toggles, implement same/cross-size switches. G11-G17 and native coordinator matrix pass.
P4 - advanced/scale and release	W6 advanced path, nightly dess-gallery stress, cargo tree feature gate, wasm size ledger, browser observations and user-facing documentation.

Non-negotiable completion conditions

- Exactly one real simulation implementation in the gallery. No parallel timer, queue interpreter or per-slice SVG rebuild.
- The old session survives all failed cross-size switches at its prior committed state; the new session becomes active only after successful publication.
- Only one pair of gallery/page lifecycle hook registrations across every scenario replacement; closures always dispatch to the current committed controller.
- Every successful slice writes the eight-cell data node once and matches oracle; a repaired or staged failed slice never masquerades as committed.
- A hidden/detached panel never advances or resumes itself; real controls keep focus, view and accessible state coherent.
- Per-commit gallery CDP runs as a separate parallel CI job, with real-wasm held callbacks and deterministic faults. Nightly stress is separate; no unsupported Firefox/Safari claim.

Appendix A. Source register and review disposition

Repository: `ChrisWhealy/svg-dom-graph` (main), inspected 10 November 2026. The existing design PDF governs engine semantics. Source lines and measured evidence must be rechecked if the implementation branch has moved.

Repository path	Contract / evidence used
design-specs/discrete_event_service_simulator.pdf	Revision 6, especially original UI section 9.
dess/s0/docs/decisions.md	Gallery predicate, crate ownership and interactive replay rules.
dess/s2/docs/surface.md; service-sim/src/projection/mod.rs	Long-lived scene, eight cells, exact ten-counter projection, deterministic writes.
dess/s3/docs/controller.md; service-sim/src/controller/mod.rs	Navigation ownership, tokens, dirty state, switch transaction and Play/Pause semantics.
service-sim/src/host.rs; service-sim/src/surface.rs	PageHost toggle, checkVisibility, observer and page lifecycle, SVG construction.
service-sim-engine/src/engine.rs; service-sim-engine/src/model.rs	Twelve counters, model fields and exact canonical state.
dess/s4/README.md; dess/s5/README.md	W5 recovery and W6 scales, evidence limitations and timer protocol.
perf/page/fault-injector.js; .github/workflows/ci.yml	Page-level DOM/timer/frame injector and cdp-build artifact consumed by proposed dess-gallery job.
demo/index.template.html; demo-app/src/lib.rs	Gallery route, lazy init, fragment registry and source frame.

Earlier review findings - retained dispositions

Review point	Resolution retained from revision 2
1 Observer target across swap	Stable weak PanelOwner and commit-time target swap - section 6.
2 Single Play/Pause	One toggle in wireframe, DOM and control contract - sections 3-4.
3 12 vs 10 vs 8 values	Exact named mapping - section 5.
4 D3 spike priority	First P0 deliverable and test protocol - section 8.
5 Coordinator module ownership	Rebuild transaction moved to service-sim, native tests - section 7.
6 Held vs real timing	Per-test lanes and one real timer smoke - sections 11-13.
7 CI cost	Single Node CDP script, G18/19 nightly - section 13.
8 Boundary attributes	aria-disabled, never disabled, click reaches controller - section 4.
9 Demo wasm cost	Build-feature audit and byte delta - section 14.
10 Catalogue labels	360-arrival scenario explicitly a scaled frozen-revision run - section 2.

Evidence correction: Revision 1 quoted 1,012 S4 failures; the newer reviewed recovery run reports 1,018. No fixed injection count is a gallery gate. This revision records no newly run gallery tests.

Revision 2 review - changes retained from revision 3

Finding	Disposition
1: faults in actual gallery	Reuse page injector; add synchronous detach-on-write mode; assert fired; one wasm build; cargo tree feature gate.
2: inert staging	Opacity:0 + inert first; off-screen + inert fallback. P0 proves presence, geometry, AX, pointer and focus.
3: replay time	Reject only by deterministic per-catalogue slice/pop ceilings and exact totals; log milliseconds as diagnostic.
4: hide/show batching	Document events must stop; same-task gallery MutationObserver batch may continue, with no double tick/leak.
5: prepare_ahead	Prohibited in ordinary gallery navigation; reserved for W6 Layer B.
6: Play at end	End of scenario status and unpressed toggle. Revision 4 requires synchronous next() at the preflight maximum instead of a transient Play boundary-probe timer.
7: range accessibility	aria-valuetext with Snapshot N of M; each arrow triggers a last-wins seek; dedupe terminal change.
8: CI placement	dess-gallery parallel job after cdp-build artifact; G18/G19 and extended G20 nightly.

Appendix B. Revision 3 review - final P0 notes

These three clarifications are normative for P0 and the gallery adapter. They do not reopen D1-D5, change the order P0-P4, alter the engine/controller algorithms or claim completed implementation.

Clarification	Required implementation contract / acceptance
R4-1: no end-of-run timer	At committed_slice == preflight maximum, the Play handler sends next(), not play(). The controller must return AtBoundary without a timer; last_slice is discovered even if it began as None. Dirty boundary repair still runs. The status reads End of scenario; the toggle remains Play and aria-pressed=false. Check both clean and dirty-final cases.
R4-2: shared injector isolation	Add detach-on-write to the existing page injector, default off and independently armed. Preserve existing __faults record/count/exception semantics and __timers/__frames. Assert exact single firing after a real native write, fail a missing firing, disarm cleanly and rerun selftest-recovery.mjs. One production gallery wasm; cargo tree feature gate still forbids fault-hook.
R4-3: opening wording	The cover disposition now states that revision 4 incorporates the three final clarifications. The garbled revision 3 opening sentence has been corrected.

P0 checks in execution order

- **P0-A.** Record browser/toolchain/build identity. In a real connected SVG test host, prove `opacity:0 + inert` gives live presence, measurable geometry, no keyboard or pointer access and AX exclusion; attempt off-screen inert only if it fails.
- **P0-B.** Extend the existing JavaScript injector under explicit arming and prove default-off behaviour and unchanged call-count traces before using it on the simulator.
- **P0-C.** Trigger synchronous mid-draw detach through an actual Surface overlay/data write; require the exact method and ordinal and a single firing; verify the controller refuses the commit and repairs only after reattachment.
- **P0-D.** Run unchanged original injector recovery self-tests, the new detach self-test and the demo feature-isolation check. Record outcomes and block P1 if any required P0 gate fails.

Evidence record and scope

Record exact CSS and inert state, before/after measurements, checkVisibility/presence, AX/focus/pointer readbacks, wrapped operation list, armed/fired report, original-vs-extended call counts, no-fire negative control, controller hashes/cursor/dirty status and browser version. Distinguish a designed gate from a newly measured result. Chrome remains the release gate; Firefox/Safari remain observations.

> **Change control:** This is a narrow clarification revision. No design changes to the scenario catalogue, deterministic replay ceilings, coordinator ownership, one-toggle control, lifecycle semantics, or per-commit/nightly CI placement.